The six trignometric ratios are

$$\sin\theta = \frac{opposite\ side}{hypotenuse} = \frac{7}{25} \qquad \cos\theta = \frac{adjacent\ side}{hypotenuse} = \frac{24}{25}$$

$$\tan\theta = \frac{opposite\ side}{adjacent\ side} = \frac{7}{24} \qquad \text{cosec}\theta = \frac{hypotenuse}{opposite\ side} = \frac{25}{7}$$

$$\sec\theta = \frac{hypotenuse}{adjacent\ side} = \frac{25}{24} \qquad \cot\theta = \frac{adjacent\ side}{opposite\ side} = \frac{24}{7}$$

Example 6.3 If $\tan A = \frac{2}{3}$, then find all the other trigonometric ratios.

Solution

$$\tan A = \frac{opposite\ side}{adjacent\ side} = \frac{2}{3}$$

By Pythagoras theorem,

$$AC = \sqrt{AB^2 + BC^2}$$

$$= \sqrt{3^2 + 2^2} = \sqrt{9+4} = \sqrt{13}$$

$$AC = \sqrt{13}$$

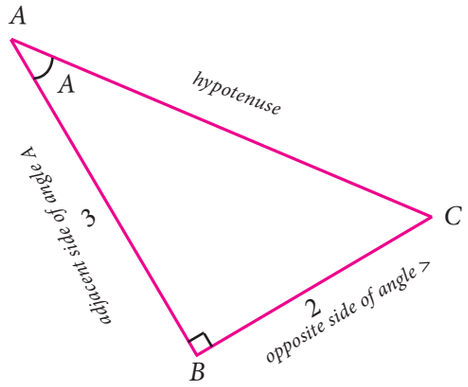


Fig. 6.7

$$\sin A = \frac{opposite\ side}{hypotenuse} = \frac{2}{\sqrt{13}} \qquad \cos A = \frac{adjacent\ side}{hypotenuse} = \frac{3}{\sqrt{13}}$$

$$\text{cosec}A = \frac{hypotenuse}{opposite\ side} = \frac{\sqrt{13}}{2} \qquad \sec A = \frac{hypotenuse}{adjacent\ side} = \frac{\sqrt{13}}{3}$$

$$\cot A = \frac{adjacent\ side}{opposite\ side} = \frac{3}{2}$$

Example 6.4 If $\sec\theta = \frac{13}{5}$, then show that $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta} = 3$

Solution:

Let $BC = 13$ and $AB = 5$

$$\sec\theta = \frac{hypotenuse}{adjacent\ side} = \frac{BC}{AB} = \frac{13}{5}$$

By the Pythagoras theorem,

$$AC = \sqrt{BC^2 - AB^2}$$

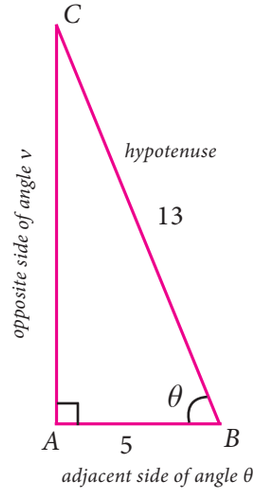


Fig. 6.8

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